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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2015 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Wednesday 5th August 2015

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 112 boys

Examiner

PKH

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following is an odd function?

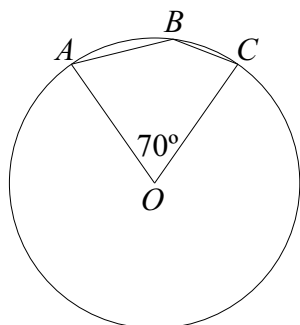
- (A) $f(x) = \tan^{-1} x$
- (B) $f(x) = \cos x$
- (C) $f(x) = \sin(x - \frac{\pi}{4})$
- (D) $f(x) = \cos^{-1} x$

QUESTION TWO

Suppose θ is the acute angle between the lines $y - 2x = 3$ and $3y = -x + 2$. Which of the following is the value of $\tan \theta$?

- (A) 7
- (B) -7
- (C) 1
- (D) -1

QUESTION THREE



What is the size of $\angle ABC$?

- (A) 110°
- (B) 145°
- (C) 140°
- (D) 130°

Exam continues next page ...

QUESTION FOUR

What is the inverse function of $f(x) = x^2 + 1$ for $x \leq 0$?

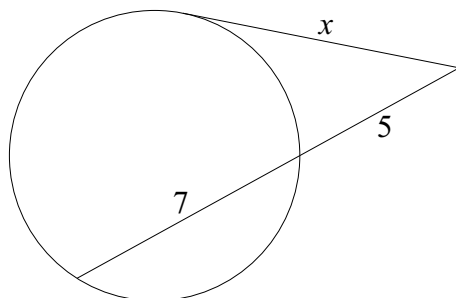
- (A) $f^{-1}(x) = -\sqrt{x-1}$, for $x \leq 0$
- (B) $f^{-1}(x) = \sqrt{x-1}$, for $x \leq 0$
- (C) $f^{-1}(x) = -\sqrt{x-1}$, for $x \geq 1$
- (D) $f^{-1}(x) = \sqrt{x-1}$, for $x \geq 1$

QUESTION FIVE

Find $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$.

- (A) ∞
- (B) $-\infty$
- (C) -5
- (D) 5

QUESTION SIX



Find the length of x .

- (A) $\sqrt{35}$
- (B) $\sqrt{12}$
- (C) $\sqrt{60}$
- (D) $\sqrt{84}$

QUESTION SEVEN

If $f(x) = \tan^{-1} \frac{1}{x}$, find $f'(x)$.

- (A) $\frac{x^2}{1+x^2}$
- (B) $-\frac{1}{1+x^2}$
- (C) $\frac{1}{1-x^2}$
- (D) $-\frac{x^2}{1-x^2}$

QUESTION EIGHT

How many solutions does the equation $x^{\frac{1}{3}} = |x-2| - 3$ have?

- (A) 0
- (B) 1
- (C) 2
- (D) 3

QUESTION NINE

The parametric form of a parabola is $(6t, -3t^2)$. Its focal length is:

- (A) $\frac{1}{4}$
- (B) $-\frac{1}{4}$
- (C) -3
- (D) 3

QUESTION TEN

The polynomial $P(x)$ has degree 4 and the polynomial $Q(x)$ has degree 2. If you divide $P(x)$ by $Q(x)$, the remainder has degree:

- (A) 1
- (B) 2
- (C) 0 or 1
- (D) 0, 1 or 2

_____ End of Section I _____

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

- (a) Let $A = (-1, 4)$ and $B = (5, -5)$. Find the co-ordinates of the point P which divides interval AB in the ratio $1 : 2$. **2**
- (b) Solve the inequation $\frac{x}{2x+1} < 2$. **2**
- (c) Sketch the graph of $y = 2 \cos^{-1}(x - 1)$, clearly marking the domain and range. **2**
- (d) Differentiate $e^{\tan x} \ln x$. **2**
- (e) Find the coefficient of a^3 in the expansion of $(2a - 1)^{20}$. **2**
- (f) Taking $x = 1.4$ as a first approximation, use one application of Newton's method to find a better approximation to $3 \sin 2x - x = 0$. **2**
Give your answer correct to 3 significant figures.
- (g) (i) Prove that $\frac{\sin 2A}{1 - \cos 2A} = \cot A$. **1**
- (ii) Hence find the values of a and b if $\cot \frac{3\pi}{8} = a + \sqrt{b}$ for integers a and b . **2**

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**

(a) Use the substitution $u = \tan x$ to evaluate $\int \frac{\sec^2 x}{\tan^2 x + 3} dx$. **2**

(b) Prove by Mathematical Induction that, for $n \geq 1$, **3**

$$\frac{1 \times 2^0}{2 \times 3} + \frac{2 \times 2^1}{3 \times 4} + \frac{3 \times 2^2}{4 \times 5} + \dots + \frac{n 2^{n-1}}{(n+1)(n+2)} = \frac{2^n}{n+2} - \frac{1}{2}.$$

(c) Find the area bounded by $y = \frac{1}{\sqrt{1-9x^2}}$, the line $x = 0$, the line $x = \frac{\sqrt{3}}{6}$ and the x -axis. **3**

(d) Consider the function $y = x^2 + \frac{16}{x}$.

(i) Find $\frac{dy}{dx}$. **1**

(ii) Find the co-ordinates of any stationary points and determine their nature. **2**

(iii) Show that there is a point of inflexion at the x -intercept. **2**

(iv) Sketch the graph $y = x^2 + \frac{16}{x}$, showing the above information. **2**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

(a) Find $\lim_{x \rightarrow 0} \frac{\sin ax}{x}$. 1

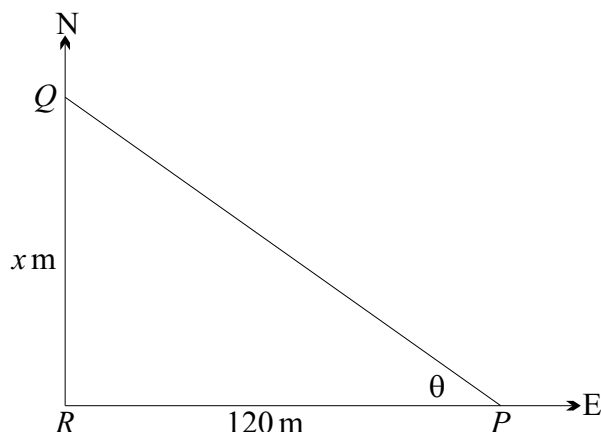
(b) (i) Show that $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$. 1

(ii) If $v^2 = 24 - 6x - 3x^2$, find the acceleration of the particle at the particle's greatest displacement from the origin. 3

(c) Let α , β and γ be the roots of the equation $x^3 - px + q = 0$. In terms of p and q find an expression for $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 2

(d) Show that $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$. 2

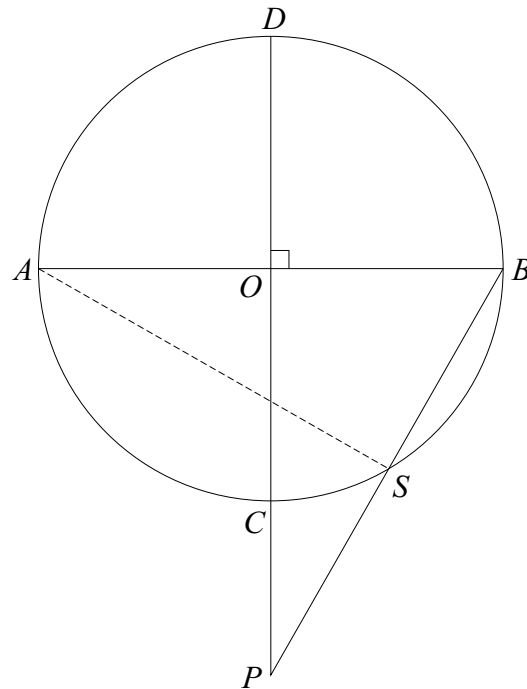
(e) 3



An observer stands at P , 120 metres East of R . A second person is at Q , x metres due North of R and continues to move North. Let angle $RPQ = \theta$. Suppose θ is changing at 0.2 radians/minute.

Find the rate at which x is changing when $x = 90$ metres.

(f)



Two diameters AB and CD of a circle, with centre O , are at right angles. Diameter DC is produced to P and PB cuts the circle again at S .

(i) Prove that $AOSP$ is a cyclic quadrilateral.

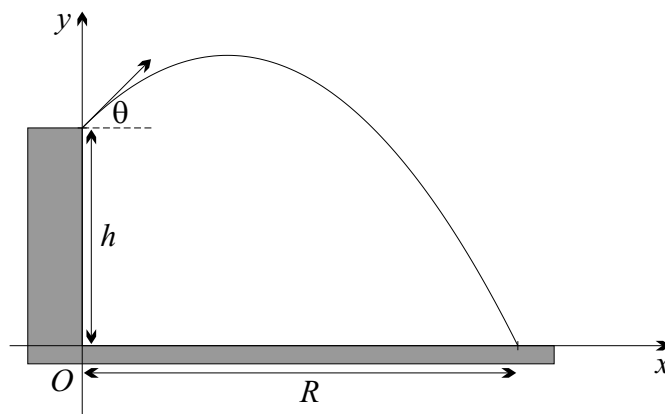
1

(ii) Prove that $\angle BCS = \angle SPO$.

2

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Consider the function $f(x) = \ln(x^{\frac{1}{x}})$, for $x > 0$.(i) Show that $f'(x) = \frac{1}{x^2}(1 - \ln x)$.**2**(ii) Find the range of $f(x)$, giving full reasons.**3**

(b)



A projectile is fired from the top of a cliff of height h above a horizontal plane with initial speed V at an angle of elevation θ . The horizontal range of the projectile is R . The magnitude of the gravitational acceleration of the projectile is g . Take the origin at the base of the cliff directly below the launch point of the projectile. It is known that the vertical and horizontal displacements satisfy

$$x = V \cos \theta t \quad \text{and} \quad y = h + V \sin \theta t - \frac{1}{2}gt^2.$$

(i) Show that the Cartesian equation of motion is

2

$$y = h + x \tan \theta - \frac{gx^2}{2V^2} \sec^2 \theta.$$

(ii) Show that $R^2 \sec^2 \theta - 2R \frac{V^2}{g} \tan \theta - 2h \frac{V^2}{g} = 0$.**2**(iii) Show that $R^2 = \left(\frac{V^4}{g^2} + 2h \frac{V^2}{g} \right) - \left(R \tan \theta - \frac{V^2}{g} \right)^2$.**2**(iv) Deduce that the maximum range is $\frac{1}{g} \sqrt{V^4 + 2hV^2g}$.**1**(v) Show that the angle of elevation satisfies $\tan \theta = \frac{V^2}{gR_1}$ where R_1 is the maximum range.**1**(vi) Show that $\tan 2\theta = \frac{R_1}{h}$.**2**

 End of Section II

END OF EXAMINATION

SOLUTION TO FORM VI SGS
TRIAL AUGUST 2015

(1)

1 (A)

2

$$y - 2x = 3$$

$$y = 2x + 3$$

$$3y = -x + 2$$

$$y = -\frac{1}{3}x + \frac{2}{3}$$

So $m_1 = 2$ and $m_2 = -\frac{1}{3}$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{2 + \frac{1}{3}}{1 - \frac{2}{3}} \right| = \left| \frac{\frac{7}{3}}{\frac{1}{3}} \right| = 7$$

(A)

3

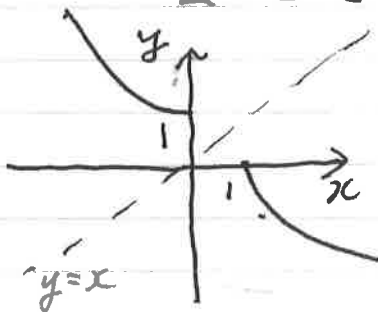
$$\angle AOC = 70^\circ$$

$$\angle AOC(\text{reflex}) = 290^\circ$$

$$\angle ABC = 145^\circ$$

(B)

4



Let $y = x^2 + 1$ $x \leq 0$
Swap x and y $y \geq 1$

$$x = \frac{y^2 + 1}{y}$$

$$y = \pm \sqrt{x-1}$$

$$f(x) = -\sqrt{x-1}, \quad x \geq 1$$

(C)

5

$$\lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{x-2}$$

$$= 5$$

(D)

2

6. $x^2 = 12 \times 5$

$x = \sqrt{60}$

(C)

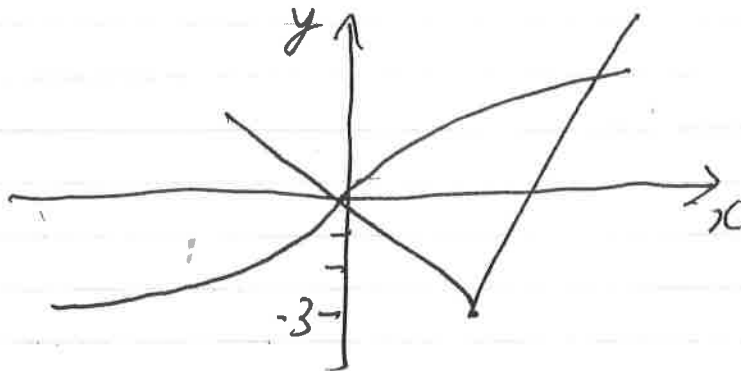
7. $y = \tan^{-1} \frac{1}{x}$

$\frac{dy}{dx} = \frac{1}{1 + \frac{1}{x^2}} \times -\frac{1}{x^2}$ (using the chain rule)

$= -\frac{1}{x^2 + 1}$

(B)

8



Two solutions

(C)

9

$P(6t, -3t^2)$

$x = 6t$

$y = -3t^2$

$t = \frac{x}{6}$

$y = -3 \left(\frac{x}{6} \right)^2$

$x^2 = -12y$

$x^2 = -4(3)y$

Focal length

is 3

(D)

10

(C)

11

3

(a) (x_1, y_1) (x_2, y_2)
 $A(-1, 4)$ $B(5, -5)$ $m:n = 1:2$

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$P = \left(\frac{1 \times 5 + 2 \times -1}{3}, \frac{1 \times -5 + 2 \times 4}{3} \right) \checkmark \checkmark$$

$$= (1, 1)$$

one mark for each co-ordinate.

(b)

$$\frac{x}{2x+1} < 2$$

$$(2x+1)x < 2(2x+1)^2$$

$$2(2x+1)^2 - (2x+1)x > 0$$

$$(2x+1)(4x+2-x) > 0$$

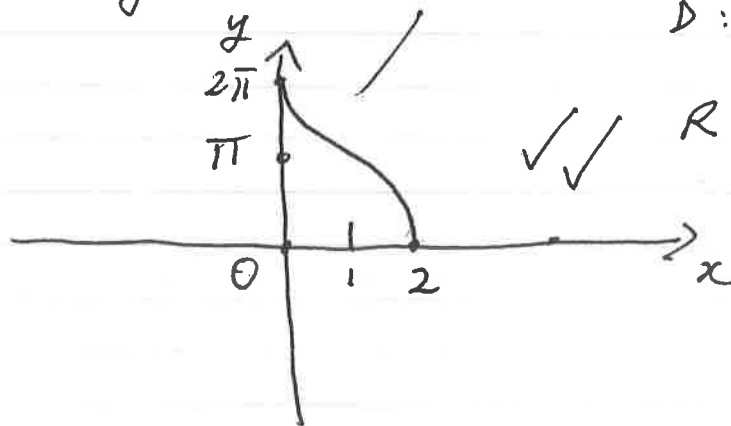
$$(2x+1)(3x+2) > 0 \quad \checkmark$$

$$x < -\frac{2}{3} \text{ or } x > -\frac{1}{2} \quad \checkmark$$



(c)

$$y = 2 \cos^{-1}(x-1)$$



$$D: -1 \leq x-1 \leq 1$$

$$0 \leq x \leq 2$$

$$R: 0 \leq y \leq 2\pi \quad \checkmark \checkmark$$

(d)

$$\text{Let } y = e^{\tan x} \ln x$$

$$\frac{dy}{dx} = e^{\tan x} \times \frac{1}{x} + e^{\tan x} \sec^2 x \ln x$$

$\checkmark$
 $\checkmark$

(4)

(e) $(2a-1)^{20}$

General term is ${}^{20}C_r (2a)^{20-r} (-1)^r$
 $= {}^{20}C_r 2^{20-r} (-1)^r a^{20-r}$ ✓

Term in a^3 is ${}^{20}C_{17} 2^3 (-1)^{17} a^3$ ✓
 $= {}^{20}C_3 2^3 \times (-1)^{17} a^3$

Coefficient is $-1 \times {}^{20}C_3 \times 8$
 $= -9120$

(f) Let $f(x) = 3 \sin 2x - x$
 $f'(x) = 6 \cos 2x - 1$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1.4 - \frac{3 \sin 2.8 - 1.4}{6 \cos 2.8 - 1}$$
 ✓
 $= 1.34$ ✓

(g) LHS = $\frac{2 \sin A \cos A}{1 - (1 - 2 \sin^2 A)}$ — *
 $= \frac{\cos A}{\sin A} = \cot A$ ✓

Using $\frac{\sin 2A}{1 - \cos 2A} = A$ with $A = \frac{3\pi}{8}$

$$\cot \frac{3\pi}{8} = \frac{\sin \frac{3\pi}{4}}{1 - \cos \frac{3\pi}{4}}$$

$$= \frac{\frac{\sqrt{2}}{2}}{1 + \frac{\sqrt{2}}{2}} = \frac{1}{\sqrt{2} + 1}$$

$= \sqrt{2} - 1$ So $\left. \begin{matrix} a = -1 \\ b = 2 \end{matrix} \right\}$ ✓

$$(a) \quad I = \int \frac{\sec^2 x}{\tan^2 x + 3} dx$$

$$\text{let } u = \tan x \\ du = \sec^2 x dx \quad \checkmark$$

$$\begin{aligned} I &= \int \frac{1}{u^2 + 3} du \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} + C \quad \checkmark \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\tan x}{\sqrt{3}} \right) \end{aligned}$$

$$(b) \quad \frac{1 \times 2^0}{2 \times 3} + \frac{2 \times 2^1}{3 \times 4} + \dots + \frac{n \times 2^{n-1}}{(n+1)(n+2)} = \frac{2^n}{n+2} - \frac{1}{2}$$

$$\text{When } n=1, \quad \text{LHS} = \frac{1 \times 2^0}{2 \times 3} = \frac{1}{6}$$

$$\text{RHS} = \frac{2^1}{3} - \frac{1}{2} = \frac{1}{6}$$

So true when $n=1$.

Assume true when $n=k$

$$\text{i.e. } \frac{1 \times 2^0}{2 \times 3} + \frac{2 \times 2^1}{3 \times 4} + \dots + \frac{k \times 2^{k-1}}{(k+1)(k+2)} = \frac{2^k}{k+2} - \frac{1}{2}$$

R.T.P true when $n=k+1$

$$\text{i.e. } \frac{1 \times 2^0}{2 \times 3} + \frac{2 \times 2^1}{3 \times 4} + \dots + \frac{k \times 2^{k-1}}{(k+1)(k+2)} + \frac{(k+1) \times 2^k}{(k+2)(k+3)} = \frac{2^{k+1}}{k+3} - \frac{1}{2}$$

$$\text{LHS} = \frac{2^k}{k+2} - \frac{1}{2} + \frac{(k+1) \times 2^k}{(k+2)(k+3)} \quad (\text{by assumption})$$

$$= \frac{(k+3)2^k + (k+1)2^k}{(k+2)(k+3)} - \frac{1}{2}$$

$$= \frac{2^k(2k+4)}{(k+2)(k+3)} = \frac{2^{k+1}}{k+3} - \frac{1}{2} = \text{RHS}$$

Hence true by Mathematical Induction

(6)

$$\begin{aligned}
 (c) \quad A &= \int_0^{\frac{\sqrt{3}}{6}} \frac{1}{\sqrt{1-(3x)^2}} dx \quad \text{obc} \\
 &= \frac{1}{3} \sin^{-1}(3x) \Big|_0^{\frac{\sqrt{3}}{6}} \quad \checkmark \\
 &= \frac{1}{3} \sin^{-1} \frac{\sqrt{3}}{2} \quad \checkmark \\
 &= \frac{1}{3} \cdot \frac{\pi}{3} \checkmark = \frac{\pi}{9}
 \end{aligned}$$

Maybe only worth 2.

$$\begin{aligned}
 (d) \quad (i) \quad y &= \frac{x^3 + 16}{x} \\
 y &= x^2 + 16x^{-1} \\
 y' &= 2x - \frac{16}{x^2} = 2x - 16x^{-2} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \text{Stat pts where } y' &= 0 \\
 2x - \frac{16}{x^2} &= 0 \\
 x^3 &= 8
 \end{aligned}$$

$$\begin{aligned}
 x &= 2 \quad y = 12 \quad \checkmark \\
 \text{Table of values for } y' & \\
 \begin{array}{c|c|c|c}
 x & 1 & 2 & 3 \\
 \hline
 y' & -14 & 0 & \frac{38}{9}
 \end{array} & \text{Min point at } (2, 12)
 \end{aligned}$$

$$(iii) \quad y'' = 2 + \frac{32}{x^3}$$

Possible point of inflexion where

$$y'' = 0$$

$$2 + \frac{32}{x^3} = 0 \quad \checkmark$$

$$x = \sqrt[3]{-16} \quad \text{which is an x-intercept}$$

(See over)

Table of values for y''

(7)

x	-3	$\sqrt[3]{-16}$	-2
y''	$\frac{22}{7}$	0	-2

When $x = -3$

$$y'' = 2 + \frac{32}{-27} = \frac{22}{27}$$

When $x = -2$

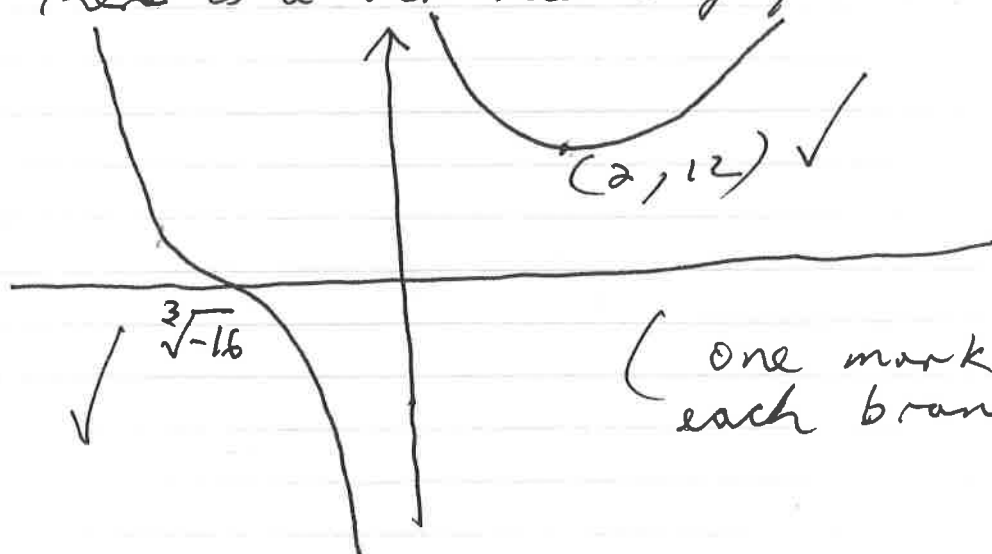
$$y'' = 2 + \frac{32}{-8} = -2$$

There is a change in concavity at $x = \sqrt[3]{-16}$

There is a point of inflexion at $x = \sqrt[3]{-16}$ (which is an x -intercept)

(iv) Now $x + \frac{16}{x} = \frac{x^3 + 16}{x}$

So there is a vertical asymptote at $x = 0$



(one mark for each branch)

$$\begin{aligned}
 (a) \quad \lim_{x \rightarrow 0} \frac{\sin ax}{x} &= a \lim_{x \rightarrow 0} \frac{\sin ax}{ax} \\
 &= a \times 1 \\
 &= a
 \end{aligned}$$

$$(b) \quad (i) \quad \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\begin{aligned}
 R.H.S. &= 1 v' \times \frac{dv}{dx} \\
 &= \frac{dv}{dt} \times \frac{dx}{dx} \\
 &= \frac{dv}{dt} = L.H.S.
 \end{aligned}$$

(ii) Particle is stationary where $v^2 = 0$.

$$24 - 6x - 3x^2 = 0$$

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0$$

Stationary at $x = -4$ and $x = 2$



$$\begin{aligned}
 \text{Now } \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \frac{d}{dx} \left(12 - 3x - \frac{3}{2} x^2 \right) \\
 &= -3 - 3x
 \end{aligned}$$

Max displacement when $x = 2$

Acceleration when $x = 2$ is

$$\begin{aligned}
 \ddot{x} &= -3 - 3 \times 2 \\
 &= -9 \text{ m/s}^2
 \end{aligned}$$

(c)

$$x^3 - px + q = 0$$

(9)

sum of roots
two at a
time

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} \quad \checkmark$$

$$= \frac{\frac{c}{a}}{-\frac{b}{a}}$$

$$= \frac{-\frac{p}{1}}{-\frac{q}{1}} = \frac{p}{q} \quad \checkmark$$

(d)

$$\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$$

$$\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$$

$$\text{Let } \alpha = \tan^{-1} 2 \text{ and } \beta = \tan^{-1} 3$$

$$\text{So } \tan \alpha = 2 \text{ and } \tan \beta = 3 \quad \checkmark$$

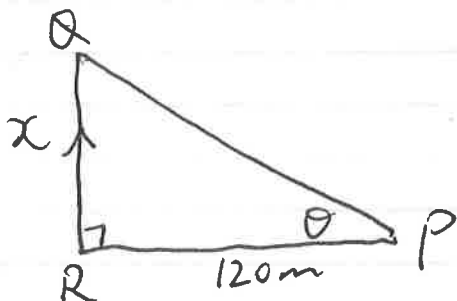
$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{2 + 3}{1 - 6} = -1$$

$$\alpha + \beta = \tan^{-1}(-1) \quad \checkmark$$

$$\text{i.e. } \alpha + \beta = \frac{\pi}{4}$$

(b)



$$\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt} \quad \checkmark$$

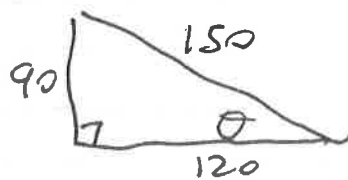
$$\begin{aligned} \frac{dx}{dt} &= 120 \sec^2 \theta \times 0.2 \\ &= 120 \times \frac{25}{16} \times 0.2 \end{aligned}$$

$$= 37.5 \text{ m/min} \quad \checkmark$$

$$\frac{x}{120} = \tan \theta$$

$$x = 120 \tan \theta$$

$$\frac{dx}{d\theta} = 120 \sec^2 \theta \quad \checkmark$$



$$\sec \theta = \frac{150}{120} = \frac{5}{4} \quad \checkmark$$

10



$$\angle ASL = 90^\circ \text{ (st line)}$$

(ii)

$$LSP0 = LSA0 (=LSAB)$$

$$\angle BCS = \angle SPO$$

15 (i) $f(x) = \ln(x^{\frac{1}{x}}) \quad x > 0$

$$f(x) = \frac{1}{x} \ln x$$

$$f'(x) = \frac{1}{x} \cdot \frac{1}{x} + \ln x \times -\frac{1}{x^2}$$

$$= \frac{1}{x^2} (1 - \ln x) \quad \checkmark$$

(ii) Stationary pts where $f'(x) = 0$

$$\frac{1}{x^2} (1 - \ln x) = 0$$

$$1 - \ln x = 0$$

$$\ln x = 1$$

$$x = e, \quad y = \frac{1}{e} \quad \checkmark$$

Table of values for y'

x	2	e	3
y'	0.08	0	-0.01



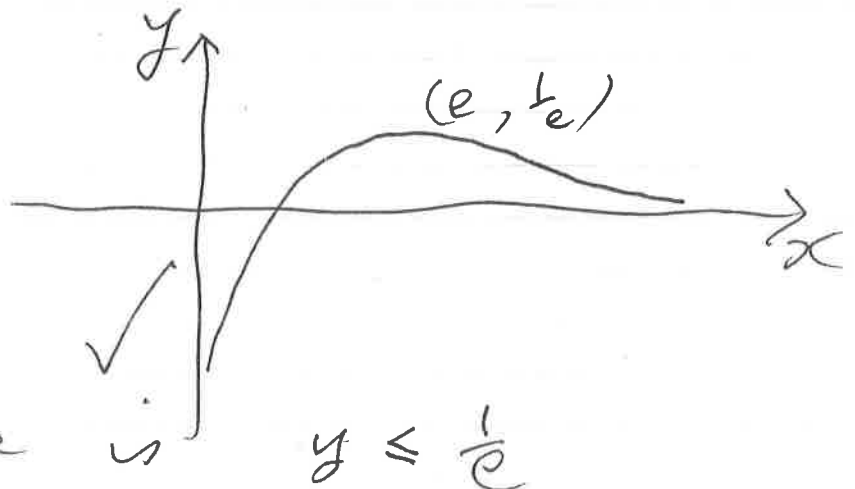
Max point at $(e, \frac{1}{e})$ $\checkmark$

Now as $x \rightarrow 0^+$, $y \rightarrow -\infty$
 $x \rightarrow \infty$, $y \rightarrow 0^+$

OR Sketch

(and correct range)

Range is $y \leq \frac{1}{e}$



QUESTION 15

(12)

$$(b) \quad x = v \cos \theta t \quad y = h + v \sin \theta t - \frac{1}{2} g t^2$$

$$(i) \quad t = \frac{x}{v \cos \theta}$$

$$y = h + v \sin \theta \cdot \frac{x}{v \cos \theta} - \frac{1}{2} g \frac{x^2}{v^2 \cos^2 \theta}$$

$$y = h + x \tan \theta - \frac{1}{2} \frac{g x^2}{v^2} \sec^2 \theta.$$

(ii) Passes through $(R, 0)$

$$0 = h + R \tan \theta - \frac{1}{2} \frac{g R^2}{v^2} \sec^2 \theta$$

$$\times \frac{2v^2}{g}: \quad R^2 \sec^2 \theta - \frac{2Rv^2}{g} \tan \theta - \frac{2h v^2}{g} = 0$$

$$(iii) \quad \text{But } \sec^2 \theta = \tan^2 \theta + 1$$

$$R^2 \tan^2 \theta + R^2 - \frac{2Rv^2}{g} \tan \theta + \frac{v^4}{g^2} - \frac{v^4}{g^2} - \frac{2h v^2}{g} = 0$$

$$R^2 + \left(\tan \theta - \frac{v^2}{g} \right)^2 - \left(\frac{v^4}{g^2} + \frac{2h v^2}{g} \right) = 0$$

$$R^2 = \left(\frac{v^4}{g^2} - \frac{2h v^2}{g} \right) - \left(\tan \theta - \frac{v^2}{g} \right)^2$$

$$(iv) \quad R^2 \leq \frac{v^4}{g^2} - \frac{2h v^2}{g}$$

$$R \leq \frac{1}{g} \sqrt{v^4 - 2h v^2 g}$$

Max R_1 occurs where

$$R_1 \tan \theta - \frac{v^2}{g} = 0$$

$$\tan \theta = \frac{v^2}{R_1 g}$$

✓

(vi)

$$R_1 = \sqrt{\frac{v^2}{g^2} \left(\frac{v^2}{g} + 2h \right)}$$

$$= \frac{v}{g} \sqrt{\frac{v^2}{g} + 2h} \quad \text{---} *$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= 2 \frac{v^2}{R_1 g} \div 1 - \frac{v^4}{R_1^2 g^2}$$

✓

$$= \frac{2 v^2}{R_1 g} * \frac{R_1^2 g^2}{R_1^2 g^2 - v^4} **$$

See over

$$R_1 = \sqrt{\frac{v^4}{g^2} + \frac{2h v^2}{g}}$$

$$R_1 = \frac{v}{g} \sqrt{v^2 + 2hg}$$

$$R_1^2 = \frac{v^2}{g^2} (v^2 + 2hg)$$

$$g^2 R_1^2 = v^4 + 2v^2 hg$$

$$g^2 R_1^2 - v^4 = 2v^2 hg$$

Substitute into

$$\begin{aligned} \tan 2\theta &= \frac{2v^2}{R_1 g} \times \frac{R_1^2 g^2}{2v^2 hg} \\ &= \frac{R_1}{h} \end{aligned}$$